

Installation Procedure

Surface preparation, membrane layout, adhesive application, perimeter detailing and hot-air welding

READ BEFORE INSTALLING: A contractor experienced with DeckRite or comparable PVC membrane systems is recommended. Read these instructions and the adhesive label completely before beginning. Site conditions and product labels govern application and safety requirements.

1. Surface Preparation

Wood Deck - Unsealed

- Provide positive drainage. A minimum slope of 1/16 inch per foot is recommended.
- For an existing deck, install 1/2" or 5/8" T&G Exterior-Grade Plywood (CDX or Better) or Exterior-rated pressure treated plywood (ACX or CDX) as a subfloor. For a new deck, install 3/4" wood option. Ensure that the plywood has the "X" grade included, which signifies exterior use glue that handles long term humidity and moisture cycles.
- Fasten plywood securely to joists or supports with corrosion-resistant screws. All fasteners must be flush with or countersunk below the surface.
- Fill surface flaws, low spots and panel joints with an appropriate patching that is rated for exterior use. Allow it to dry fully, then sand the surface smooth. Use just enough to fill the voids in the seams of the plywood. Excessive amounts will cause unnecessary sanding, drying time, and could impact adhesive effectiveness.
- Remove dust, debris, moisture, oil and other contaminants before applying adhesive.
- Do not seal plywood when using water-based adhesive. Water-based adhesives require a porous surface. If the wood is sealed or nonporous, use the approved PVC solvent-based contact adhesive.
- Note: If you are using engineered wood like AdvanTech or similar product that are rated as Exposure 1 (Designed to resist the elements and stay structurally stable during temporary construction delays only), and/or has a slightly textured surface, it is recommended that a thin layer (typically 1/4" or 3/8") of sanded exterior-grade ACX plywood be fastened over the AdvanTech or similar product to ensure adhesive compatibility and smooth installation.

Concrete / Sealed Wood Deck

- Concrete or sealed wood must be clean, smooth, dry and sound. Repair cracks larger than hairline and level uneven areas with a compatible concrete patching product; allow repairs to cure fully.
- Remove dust and surface contamination before installation.
- Concrete applications require the approved PVC **solvent-based contact adhesive**, applied to both the substrate and the back of the membrane.

2. Plan the Membrane Layout

- DeckRite membrane is supplied in 68-inch-wide rolls. Confirm all field measurements before cutting.
- Run seams parallel to the longest deck dimension and perpendicular to the direction of slope.
- Install sheets from the low point toward the high point so overlaps shed water like shingles.
- Provide at least **1 inch of overlap** at field seams for hot-air welding. Keep the overlap completely free of adhesive and contamination.
- Allow 4 to 6 inches of extra membrane at walls and outer edges for termination. Some details may require more.

LAYOUT TIP: Taking time to dry-fit, align and mark each sheet before adhesive is applied will reduce waste and prevent alignment problems.

Adhesive Selection and Application

Use only adhesive approved for DeckRite PVC membrane. The adhesive container label and current Safety Data Sheet take precedence where requirements differ.

Requirement	Water-Based Adhesive	Solvent-Based Contact Adhesive
Use on	Porous, unsealed wood only	Concrete, vertical surfaces, sealed wood and other approved nonporous surfaces
Application	One surface: substrate	Two surfaces: substrate and vinyl backing
Roller	9 inch medium-nap roller; smaller roller for tight areas	Small-nap roller; 3-inch roller for tight areas
Coverage	Approx. 125 sq. ft. per gallon	Approx. 50-60 sq. ft. per gallon
Bonding window	Install membrane while adhesive is wet	Allow 5-10 minutes; bond when slightly tacky, with no strings
Temperature	Best if used when temperatures are 45° and rising or a heated enclosure is being used	Best if used when temperatures are 45° and rising or a heated enclosure is being used

3. Water-Based Adhesive for Porous Unsealed Wood

IMPORTANT: Water-based adhesive must be applied to porous, unsealed plywood. If the plywood has been treated with a waterproofing sealer or the surface will not absorb water, use the approved solvent-based adhesive.

- After dry-fitting the sheet, place weighted objects near one end. From the opposite end, loosely roll the membrane back toward the weights while keeping the roll straight.
- Before opening, gently shake or mix the adhesive for approximately **60 seconds**. It should resemble thick white paint.
- Working in manageable sections, pour adhesive in a serpentine pattern and spread it evenly with a **medium-nap roller**. If working alone, limit each section to approximately **15 linear feet**.
- Apply a uniform coat comparable to a heavy coat of paint. Cover the surface completely, but do not leave puddles, ridges or dry spots.
- Immediately unroll the membrane into the **wet adhesive**. Keep the sheet straight and wrinkle-free. If necessary, lift the membrane while the adhesive is still wet, reposition it and lay it back down.
- Use a clean, stiff-bristled push broom or linoleum roller with medium pressure to remove trapped air, distribute adhesive and create continuous contact.
- Continue in sections. Slightly overlap each new adhesive application with the preceding wet section so there are no gaps in coverage.
- Keep adhesive out of every seam-welding area. Remove fresh water-based adhesive from the face of the membrane immediately with a clean damp rag.

Drying and Return to Service

- Light walking is possible immediately, but excessive traffic can create wrinkles while the adhesive remains wet.
- In dry conditions between 60°F - 90°F, adhesive typically dries in 2 to 3 hours. Allow it to dry overnight before installing equipment or placing concentrated loads whenever possible.

Solvent-Based Adhesive and Membrane Placement

For concrete, vertical surfaces, sealed wood and other approved nonporous substrates

4. Solvent-Based Contact Adhesive

SAFETY: Solvent-based adhesive is highly flammable. Keep it away from children, sparks, open flame and high heat. Work only in a well-ventilated area and follow the adhesive label and Safety Data Sheet for required personal protection.

- Dry-fit and pre-cut the membrane where practical. Contact adhesive bonds quickly, so confirm the alignment before the glued surfaces touch.
- With a small-nap roller, apply a thin, uniform coat to the vinyl backing of the membrane. Do not allow adhesive to contact the finished face.
- Apply a thin, uniform coat to the receiving substrate. Coat only the area that can be covered within the working time.
- Allow both coated surfaces to set for approximately 5 - 10 minutes, depending on temperature, humidity and air movement.
- Test the adhesive on the substrate with a clean finger. It is ready when it feels only slightly tacky. If the adhesive forms legs or strings, wait another 5 minutes and test again.
- Do not wait longer than the adhesive label allows. Carefully position and bond the membrane, then smooth it firmly by hand or roller to remove air and ensure uniform contact.
- Approximate coverage is 50 to 60 sq. ft. per gallon because adhesive is applied to both surfaces. Actual coverage varies with porosity, roller nap and application thickness.

Temporary Gas Bubbles

- Applying solvent-based adhesive too heavily may produce temporary gas pockets beneath the membrane after approximately 24 hours.
- Do not remove the membrane or seal the edges while the gases are escaping. If practical, warm the installation in sunlight. The membrane should draw down as the solvent dissipates; this may take 2 to 3 days depending on conditions and adhesive thickness.

Drying and Return to Service

- Solvent-based contact adhesive dries quickly. Allow at least 1 hour before installing equipment or placing concentrated loads, unless the product label requires longer.

5. Complete the Membrane Installation

- Before gluing, sweep or blow the deck clean and pre-cut around posts or penetrations where required.
- Use tape-measure marks or a lightly snapped white chalk line about every 4 feet to maintain alignment. Excess chalk can contaminate a weld area.
- Roll out the first sheet along the guide marks and verify the required extra material at walls and deck edges.
- Fold or roll back a manageable portion, apply the selected adhesive according to the appropriate procedure, and place the membrane without trapping air.
- Continue until the entire sheet is bonded. Repeat for remaining sheets while preserving clean, adhesive-free seam overlaps.

DO NOT TRACK ADHESIVE: Avoid stepping in adhesive and walking onto the finished membrane. Solvent-based adhesive is extremely difficult to remove from the face of the membrane.

Welding Perimeter Details and Job Checklist

Complete all details in accordance with current DeckRite requirements and applicable building codes.

6. Hot-Air Welding

- Use an approved hot-air welder, such as Leister or Steinel, fitted with a **40 mm nozzle**, together with a **1.5 inch silicone roller**.
- Allow the welder to warm up. Begin with the rheostat near setting 7, then adjust for tool model, ambient conditions and welding speed.
- Always test-weld scrap membrane first. The membrane weld range is approximately **450 to 500 degrees F**, but a proper weld depends on the combination of heat, pressure and travel speed.
- Insert the nozzle between the overlapping sheets, leaving about 1/8 inch of the nozzle visible. Draw the welder steadily toward you while following immediately with the silicone roller held parallel to the nozzle.
- Inspect and probe the cooled seam to confirm a continuous weld before completing perimeter details.

7. Perimeter and Wall Termination

- Secure pre-cut DeckRite termination bar around the deck perimeter with the caulk lip facing down and away from the deck. Fasten with approved color-coated screws or masonry tappets.
- Approved PVC coated metal drip edge may also be used. Follow the manufacturer's current installation instructions for the selected perimeter system.
- At walls, the membrane may be turned up behind siding and sealed with a continuous bead of compatible caulk, or incorporated into an approved flashing detail.
- Contact DeckRite for project-specific termination and flashing guidance.

Suggested Tools	DeckRite Materials	Other Materials
Utility knife Scissors Measuring tape White chalk line Straightedge Stiff-bristled push broom 9 inch medium-nap roller Small-nap roller 1.5 inch silicone roller Hot-air welding gun Drill or screwdriver Caulk gun Extension cord Clean rags	DeckRite membrane Approved water-based adhesive Approved solvent-based adhesive Compatible caulk Termination bar Color-coated screws	Min. BCX grade pressure-treated plywood Corrosion-resistant screws Concrete patching material Compatible joint or patching compound Approved coated metal drip edge, if used

TECHNICAL ASSISTANCE: Call DeckRite at 888-450-DECK (3325) or contact your independent DeckRite representative before proceeding if site conditions differ from these instructions.